A project proposal for partial fulfilment of the course unit IT2153- Group Project for the degree of Information Technology

AI-Powered Document Summarizer

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Declaration

We hereby declare that the project proposal submitted for evaluation of course module IT2153 leading to the award of a Bachelor of Science in Information Technology is entirely our own work, and the contents taken from the work of others have been cited and acknowledged within the text. This proposal has not been submitted for any degree at this University or any other institution.

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I recommend the project to be carried out by the students,

_____ _____

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Contents

1. Introduction 4
1.1. Introduction 4
1.2. Objectives 4
1.3. Befits of this research 4
2. Background 5
2.1. Background 5
2.2. Review on the existing systems. 5
3. Material and Methods 5
4. Expected Results 6
5. Timeline of the Research 6
6. References 8

AI-Powered Document Summarizer

1. Introduction

1.1. Introduction

With the rapid increase in digital document usage, there is a significant need for tools that can efficiently summarize and extract key information from large volumes of text. This project aims to develop an AI-powered document summarizer that accepts image or PDF inputs, processes them, and provides concise summaries to users.

1.2. Objectives

1. To develop an application that allows users to upload images and PDFs for summarization.
2. To utilize machine learning algorithms to extract and summarize text effectively.
3. To implement a user-friendly interface using C# for seamless interaction.
4. To ensure accurate summarization using natural language processing techniques in Python.

1.3. Befits of this research

Time-saving: Automates the summarization process, reducing the time users spend reading long documents.

Enhanced comprehension: Provides clear and concise summaries, improving user understanding of key concepts.

Accessibility: Makes information retrieval straightforward and efficient for various users.

2. Background

2.1. Background

The growing amount of information available in digital formats presents challenges in managing and interpreting data. Traditional methods of summarization are often manual and time-consuming. This project seeks to leverage AI technology to create a solution that automates this process, enhancing productivity in academic, professional, and personal contexts.

2.2. Review on the existing systems.

Current document summarization tools often focus on text documents and lack support for image and PDF formats. Additionally, many tools provide basic summarization without leveraging advanced machine learning techniques. This project will build upon existing solutions by integrating robust summarization algorithms and supporting a wider range of input formats.

3. Material and Methods

Programming Languages: C# for the front-end application, Python for backend processing and summarization, and SQL for database management.

Technologies:

- Optical Character Recognition (OCR) for extracting text from images and PDFs.
- Natural Language Processing (NLP) libraries (e.g., NLTK, SpaCy) for summarization.
- SQL database to store user inputs and generated summaries.

The proposed system will consist of:

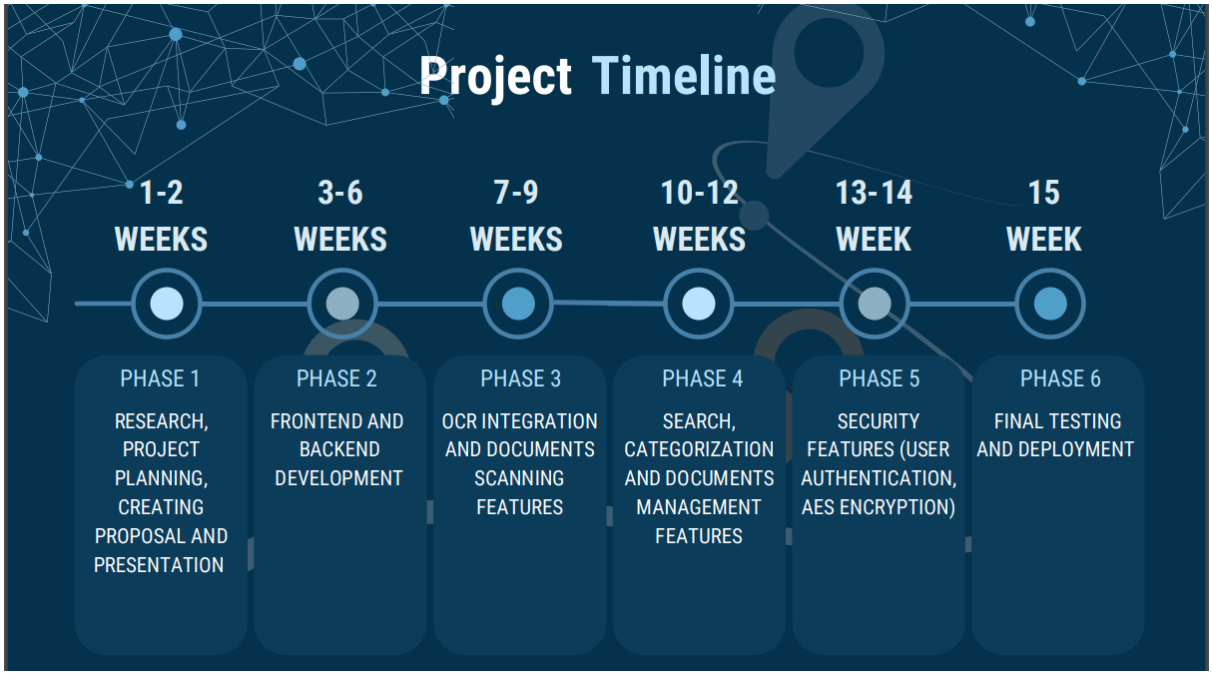
1. **Front-end Application**: Developed in C#, this application will allow users to upload documents (images/PDFs) and display generated summaries.
2. **Backend Processing**: A Python service will handle OCR and NLP processes to extract and summarize text.

3. **Database**: A SQL database will store uploaded documents and their corresponding summaries for future reference.

4. Expected Results

- **Physical Document uploading Technology :**
  Real time Scanning + already saved file Uploading + through website links
- **Uploaded document Enhancing :**
  Light , edges editing ,sharpness editing(Real-Time scanned)
  Selecting No.of pages in the doc , crop etc(External uploads)
- **Then OCR process**
- **Features** :
  -Summarization
  -convert into(pdf, doc etc)
  -Save(Local + Cloud)
  -Share
  -categorization
- -Security (Password , user authentic login ,etc)

5. Timeline of the Research



Gantt chat

Search, Categorization,	Document Scanning Features.	OCR Integration	Backend Development	Frontend Developmen	Proposal and Presentation.	Research, Planning	Task
2 Weeks	2 Weeks	2 Weeks	4 Weeks	4 Weeks	2 Weeks	2 Weeks	Duration
					×	×	Week 1
					×	×	Week 2
			×	×			Week 3
			×	×			Week 4
			×	×			Week 5
			×	×			Week 6
	×	×					Week 7
	×	×					Week 8
	×	×					Week 9
×							Week 10
×							Week 11
×							Week 12
							Week 13
							Week 14
							Week 15

Document Management	2 Weeks										x	x	x			
Security Features	1 Weeks													x	x	
Final Testing and Deployment.	1 Weeks															x

X indicates the weeks allocated for each task.

6. References

1. Vaswani, A. et al. (2017). "Attention is All You Need."
2. Goodfellow, I., Bengio, Y., & Courville, A. (2016). "Deep Learning."
3. Jha, S. et al. (2020). "Automating Document Summarization Using Deep Learning Techniques."